

KPI ADVISORY & ANALYTICS

Functional Specification Document

A unified blueprint translating business strategy into governed, measurable performance metrics.

Document Control & Metadata

Document Version	1.0
Generated Date	June 09, 2026
Industry Sector	Manufacturing
Organizational Level	CXO
Scope of Performance Metrics	1 Approved Performance Metrics
Approval Status	DRAFT

1. Executive Summary

This document outlines the functional specifications for the approved performance metrics of the organization. Each metric is mapped to strategic objectives and key result areas to ensure operational alignment and governance.

2. KPI Landscape Overview

The following table provides a high-level catalog of all approved performance indicators within the scope of this transformation initiative.

KPI ID	KPI Name	Category	Functional Area
KPI-001	Cost Per Unit	Operational	Operations

3. Strategic Traceability Matrix

This matrix illustrates the strategic alignment from executive objectives down to specific key performance indicators, providing visibility into strategic translation.

Strategic Objective	Business Challenge	KRA	Functional Area	KPI Name
Improve Gross Margin	High Operational Cost	Operational Excellence	Operations	Cost Per Unit

4. Individual KPI Specifications

Metric 01: Cost Per Unit

KPI Category	Operational	Functional Area	Operations
Business Owner	Director of Operations	Data Owner	Finance Manager

Business Definition

Measures and evaluates the efficiency of Cost Per Unit against design parameters to ensure stable throughput.

Business Purpose & Strategic Relevance

Why it exists: Sourced directly to support strategic tracking of Operational Excellence. Business value: Enables the steering committee to monitor process bottlenecks and drive resource efficiency. Decisions supported: Operational resource planning, shift schedules, and capacity allocations. Risks of not monitoring: Operational margin leakage and process variability.

Calculation Methodology

Formula	Total Cost / Total Units
Numerator Details	Total Cost
Denominator Details	Total Units
Calculation Logic	1. Extract the raw transaction history from SAP. 2. Filter out non-conforming test runs and quality checkpoints. 3. Sum the daily output for the numerator. 4. Divide by the total inputs in the denominator, then multiply by 100 for the percentage score.
Inclusion Rules	Include all validated customer orders and completed production lots.
Exclusion Rules	Exclude raw material waste, testing scrap, and canceled orders.
Sample Worked Example	If good units = 9,500 and total started = 10,000, then KPI = (9,500 / 10,000) * 100 = 95.00%.

Business Rules & Data Validation

Business Operating Rules	Standard monthly posting dates apply. Manual overrides require VP approval.
Data Validation Rules	KPI must lie in the range [0, 100]. Denominator must be strictly positive.
Exception Handling Rules	In case of zero transactions, default the metric score to Null to avoid division by zero errors.
Data Quality Expectations	Timeliness: Data loaded within 24 hours of period close. Completeness: >99.9%.

Recommended Source Systems

Source system: SAP (Module: CO). Dependencies: Master transactional tables and inventory logs. Upstream: ERP transaction ledger. Downstream: C-suite dashboard.

5. Governance Framework

To ensure metric consistency, accountability, and ongoing relevance, a formal governance framework is established for all approved KPIs. This structure assigns clear responsibilities and defines escalation paths.

Roles and Responsibilities

- **Business Owner:** Responsible for defining the business logic, validating calculation results, approving target thresholds, and driving operational performance based on metric insights.
- **Data Owner:** Accountable for technical lineage, data completeness, source-to-target mapping, ETL data quality checks, and resolving data ingestion or availability issues.
- **Escalation Path:** In case of data quality discrepancies or alignment disputes, issues are escalated to the Data Governance Committee and KPI Advisory Board for review and reconciliation.

Ownership and Governance Policy Matrix

Metric Name	Business Owner	Data Owner	Governance Policy Notes
Cost Per Unit	Director of Operations	Finance Manager	Business Owner (Director of Operations): Reviews variance and coordinates corrective actions. Data Owner (Finance Manager): Maintains data integration pipeline and semantic views. Governance: Subject to monthly audits by the data steering committee.

6. Reporting & Dashboard Requirements

Visualization layout and threshold criteria dictate how data is displayed to support decision-making. The matrix below outlines reporting recommendations and performance thresholds for each metric.

Metric Name	Reporting Guidelines	Dashboard Placement	Threshold Guidance
Cost Per Unit	Visual layout: Gauge card and trend line showing actuals vs target of \$10 - \$12.	Include on the Executive Performance Dashboard as a primary card.	Green: >= Target, Amber: Within 5% of target, Red: Below target (> \$15).

7. Assumptions & Constraints

A clear understanding of business and technical assumptions is critical for successful implementation. The following list represents consolidated baseline assumptions and constraints.

General Architectural Assumptions

- **Data Availability:** Source transactional tables are assumed to be loaded into the central reporting repository on a standard nightly batch cadence.
- **Fiscal Calendar:** Standard calendar year rules are assumed unless otherwise explicitly documented in specific financial indicators.
- **System Uptime:** Target source ERP ledgers are assumed to maintain 99.5% uptime during standard reporting extraction windows.

Metric-Specific Assumptions and Limitations

- **Cost Per Unit:** Assumptions: Continuous data stream availability and stable ERP schemas. Constraints: Batch processing latency may delay daily refresh.

8. Implementation Considerations

Transitioning these specifications into functional BI tools requires rigorous testing, change management, and data reconciliation. Standard implementation considerations are outlined below.

Standard Implementation Guidelines

- **Data Reconciliation:** All KPI calculation outcomes must be audited and reconciled against official books of record or audited financial statements.
- **User Acceptance Testing (UAT):** Business owners must perform visual and numeric validation of dashboard mockups prior to production sign-off.
- **Change Management:** Training sessions and clear system documentation are required to support user onboarding and ensure high organizational adoption.

Metric-Specific Technical Considerations

- **Cost Per Unit:** Integration: Sourced via standard API endpoints. Technical Risks: Sync latency between master replica database. Change Management: Training workshop for operations managers.

9. Appendix

KPI Glossary

- **KPI (Key Performance Indicator):** A quantifiable measure used to evaluate the success of an organization or activity in meeting performance objectives.
- **KRA (Key Result Area):** Primary focus areas of outcomes or outputs for which an organizational unit or role is responsible.
- **Numerator:** The upper portion of a division representing the measured subset of occurrences.
- **Denominator:** The lower portion of a division representing the total base population.

Data Quality Principles

- **Accuracy:** Data correctly represents the real-world operational event it records.
- **Completeness:** All necessary dataset components are present without omission.
- **Consistency:** Metrics align across various systems, business units, and report interfaces.
- **Timeliness:** Updates occur within the required reporting cadence and operational windows.

Acronym Reference

- **ERP:** Enterprise Resource Planning
- **SAP FI-CO:** Financial Accounting & Controlling
- **SAP SD:** Sales & Distribution
- **SAP MM:** Materials Management
- **BI / DWH:** Business Intelligence / Data Warehouse
- **UAT:** User Acceptance Testing